Mini Project Assessment

Business Case

You are working as a **Data Engineer** for **QuickCart**, a rapidly growing e-commerce and logistics platform.

The company currently faces:
- Slow dashboard performance
- Duplicate customer records
- Delayed reporting
- Pipeline failures
- High storage costs
- Difficulty processing clickstream logs
- Poor query performance on large datasets

The organization wants to build a scalable modern data platform using:
Python, SQL, Pandas, PostgreSQL, Airflow, dbt, Parquet, Partitioned datasets, and modern ETL/ELT architecture.

Assessment Structure

Section	Topic	Type	Marks	Duration
A	Data Architecture & Theory	Subjective	15	2 Hours
B	SQL Engineering	Practical	25	4 Hours
C	Python + Pandas Data Processing	Practical	20	3 Hours
D	ETL/ELT + Airflow + dbt	Practical	25	5 Hours
E	Optimization + Performance Engineering	Case Study	15	2 Hours
TOTAL			100	16 Hours

Expected Deliverables

Students must submit:

1.SQL scripts (.sql)
2. Python notebooks/scripts (.py /.ipynb)
3. Airflow DAG (files/scripts)
4.dbt models (files/scripts)
5.Architecture diagram (word file/paint/onenote etc.)
6.Optimization explanations

DATASETS PROVIDED

1. customers.csv
customer_id
customer_name
email
city
signup_date

2. orders.csv
order_id
customer_id
product_id
order_date
quantity
unit_price
payment_status

3. products.csv
product_id
product_name
category
supplier_id
cost_price
selling_price

4. clickstream.csv
event_id
customer_id
event_type
page_url
event_timestamp
device_type

Dataset URL : https://www.kaggle.com/datasets/wafaaelhusseini/e-commerce-transactions-clickstream

Note: On a public platform providing data, the columns could be updated or deleted. In case it hinders the questions given, kindly adjust it to the column_name given in the dataset or omit that question.

Section A - Data Engineering Theory

Q1. OLTP vs OLAP Architecture

The company currently stores transactional and analytical workloads in the same PostgreSQL database.

Tasks

1. Explain why this creates performance issues.

2. Compare OLTP and OLAP workloads.

3. Suggest an improved architecture.

4. Explain where:
   -PostgreSQL /SQLServer
   -Data warehouse
   -Data lake

should be used.

Q2. ACID vs BASE in Distributed Systems

QuickCart wants to store clickstream events from millions of users.

Tasks

1. Explain why BASE systems are preferred for clickstream systems.

2. Which workloads require ACID compliance?

3. Give examples where eventual consistency is acceptable.

Q3. Storage Formats & Querying

Tasks

1. Compare:
   -CSV
   -JSON
   -Avro
   -Parquet

2. Why do analytical systems prefer columnar storage?

3. Explain predicate pushdown.

4. Explain compression benefits in Parquet.

Q4. Partitioning, Indexing & Sharding

Tasks

1. Design a partitioning strategy for:
   -orders table
   -clickstream data

2. Suggest indexing strategy for:
   -customer lookups
   -order filtering
   -joins

3. Explain when indexes negatively affect performance.

4. Explain horizontal sharding.

Section B - SQL Engineering

Q1. Intermediate SQL Queries

Write SQL queries to:

1. Find top 10 customers by revenue.
2. Find month-over-month sales growth.
3. Find customers who ordered in consecutive months.
4. Find products never ordered.
5. Find revenue contribution percentage by category.

Q2. Advanced SQL

Tasks

1. Rank customers based on total revenue.
2. Find running total sales by month.
3. Find highest selling product per category.
4. Find 7-day rolling average sales.

Q3. SQL Optimization Scenario

The following query takes 18 minutes:

```
SELECT c.customer_name,
    SUM(o.quantity * o.unit_price)
FROM orders o
JOIN customers c
ON o.customer_id = c.customer_id
```

```
WHERE order_date >= '2025-01-01'
GROUP BY c.customer_name;
```

Tasks

1. Identify possible bottlenecks.
2. Suggest indexes.
3. Explain execution plan analysis.
4. Suggest partitioning improvements.
5. Explain materialized views usefulness.

Q4. Database Design

Design schema for:

1. Shipment tracking
2. Product inventory
3. Customer support tickets

Requirements:

- normalized design
- primary keys
- foreign keys
- indexing recommendations

Section C - Python + Pandas Engineering

Q1. Data Ingestion Pipeline

Using Python and Pandas:

Tasks

1. Read all datasets.
2. Validate schema consistency.
3. Detect duplicate customers.

4. Identify invalid records.

Q2. Data Cleaning & Transformation

Tasks

1. Standardize city names.
2. Convert timestamps correctly.
3. Handle missing values.
4. Remove corrupted rows.

Q3. Business KPI Generation

Generate:

1. Daily sales KPI
2. Customer lifetime value
3. Category-wise revenue
4. Repeat customer percentage

Q4. Clickstream Analytics

Using clickstream data:

Tasks

1. Find most visited pages.
2. Calculate session counts.
3. Find bounce rate.
4. Find mobile vs desktop traffic percentage.

Q5. Export Optimization

Tasks

1. Export analytical dataset into:
    - CSV
    - Parquet

2. Compare storage sizes.
3. Compare read performance.
4. Explain why Parquet is better for analytics.

Section D - ETL/ELT + Airflow + dbt

Q1. ETL vs ELT Architecture

Tasks

1. Compare ETL vs ELT.
2. Explain why cloud warehouses prefer ELT.
3. Design modern ELT pipeline for QuickCart.

Q2. Airflow DAG Development

Create an Airflow DAG that:

Requirements

1. Runs daily at 2 AM
2. Extracts CSV files
3. Performs validation
4. Loads raw data into staging tables
5. Triggers transformation step
6. Sends failure email alerts

DAG must include:

- retries
- retry delays
- task dependencies
- logging
- scheduling
- sensors

Q3. Airflow Failure Handling

Task

1. Explain idempotency in pipelines.

2. Explain retry mechanisms.
3. How would you prevent duplicate loads?
4. Explain backfilling.

Q4. dbt Modeling

Tasks

Create:

1. staging model
2. intermediate model
3. mart/fact model

Include:

- dbt tests
- source freshness
- incremental model
- documentation

Q5. Data Warehouse Modeling

Tasks

1. Design star schema.
2. Identify:
   fact tables
   dimension tables

Section E - Performance Engineering Case Study

Scenario

QuickCart processes:

- 50 million clickstream events/day
- 5 million orders/day

Problems:

- SQL queries are slow
- Pipelines fail frequently
- Dashboard refresh takes 3 hours
- Storage cost is increasing

Tasks

Part 1 - Architecture Design

Design scalable architecture using:

- database
- data lake
- warehouse
- orchestration
- transformation layer

Part 2 - Optimization

Explain:

1. Partitioning strategy
2. Indexing strategy
3. Compression strategy
4. Query optimization
5. Incremental loading
6. Columnar storage benefits

Part 3 - Reliability

Explain:

1. Retry handling
2. Failure alerts
3. Data quality validation
4. Monitoring strategy
5. Idempotent design